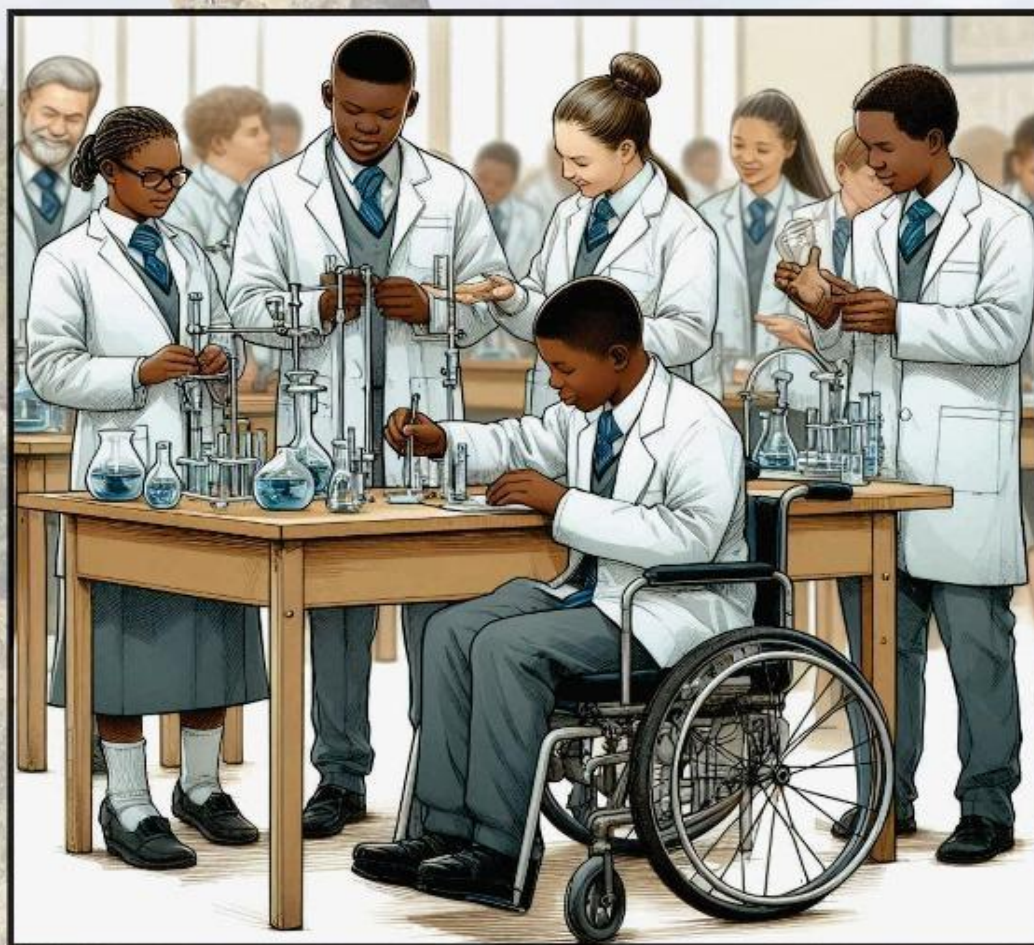




ZIMBABWE

Ministry of Primary and Secondary Education



COMBINED SCIENCE SYLLABUS

2024-2030

FORM 1 - 4

Curriculum Development and Technical Services
Box MP 133
Mt Pleasant
Harare

@ 2024
All rights reserved

ACKNOWLEDGEMENTS

The Ministry of Primary and Secondary Education would like to acknowledge contributions made by the following towards the development and production of this Heritage-based Combined Science syllabus:

- The National Science Syllabus Panel
- Zimbabwe School Examinations Council (ZIMSEC)
- Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development (MoHTEISTD)
- UNICEF
- UNESCO

TABLE OF CONTENTS

ACKNOWLEDGEMENTS	i
1.0 PREAMBLE.....	1
1.1 INTRODUCTION	1
1.2 RATIONALE	1
1.3 SUMMARY OF CONTENT	1
1.4 ASSUMPTIONS.....	1
1.5 CROSS- CUTTING THEMES	2
2.0 PRESENTATION OF THE SYLLABUS	2
3.0 AIMS	2
4.0 OBJECTIVES.....	2
5.0 TOPICS	3
6.0 METHODOLOGY AND TIME ALLOCATION	4
6.1 METHODOLOGY	4
6.2 TIME ALLOCATION.....	5
7.0 SCOPE AND SEQUENCE	6
8.0 COMPETENCY MATRIX	13
FORM 1.....	13
FORM 2.....	25
FORM 3.....	37
FORM 4.....	66
9.0 ASSESSMENT	84
9.1 ASSESSMENT OBJECTIVES	84
9.2 Assessment Model.....	85
9.3 Scheme of Assessment	86
9.4 Specification Grid	89

1.0 PREAMBLE

INTRODUCTION

This Heritage-Based Combined Science syllabus covers four years of Secondary Education, Forms 1 – 4. The syllabus provides knowledge and skills in Combined Science and a suitable preparation for the study of science beyond Ordinary level. It aims to equip learners in their diverse needs with scientific skills of long-term value in an increasingly technological world. A learner- centred and hands on approach to the learning area is adopted to develop scientific thinking and application of acquired knowledge and skills through exploitation of country's heritage.

RATIONALE

This syllabus develops learners' scientific skills in Physics, Chemistry and Biology. It develops knowledge of scientific concepts and principles and their application in solving everyday problems. It fosters creativity, innovation, critical thinking and problem-solving abilities among learners. It also enables learners to acquire 21st century scientific skills for an innovation and knowledge driven education.

SUMMARY OF CONTENT

The Combined Science learning area covers concepts in Biology, Chemistry and Physics. The syllabus covers scientific skills such as observing, recording, measuring, presentation, analysis and interpretation of data. The Combined Science also promotes the impartation of practical skills such as handling of apparatus, chemicals, plant and animal specimens safely and confidently.

1.4 ASSUMPTIONS

It is assumed that learners have:

- knowledge of the content of the Science and Technology syllabus offered in the Junior School Module
- engaged in science experiments
- engaged in project and collaborative work
- used measuring instruments such as rulers, thermometers, clocks and balances

1.5 CROSS- CUTTING THEMES

The following cross-cutting priorities have to be taken into consideration in the teaching and learning of Heritage-based Combined Science, for further studies, life and work.

- Gender stereotyping
- Children's rights and responsibilities
- Disaster risk management
- Entrepreneurship
- Health and wellbeing
- Environmental management
- Climate change

2.0 PRESENTATION OF THE SYLLABUS

This Form 1 to 4 Heritage-based Combined Science syllabus is presented as a single document which consists of the preamble, rationale, summary of content, assumptions, cross cutting themes, aims, objectives, topics, methodology, time allocation, scope and sequence, content matrix and assessment. The content is divided into Biology, Chemistry and Physics sections.

3.0 AIMS

The aims of the syllabus are to enable learners to further:

3.1 develop desirable scientific literacy

3.2 acquire critical thinking, creativity and problem-solving skills that apply to real life situations

3.3 gain scientific practical skills, accuracy, objectivity, integrity, enquiry and team work

3.4 develop attitudes relevant to science such as self-initiative, self-managing and enterprising

3.5 relate scientific practices to sustainable use of natural resource

3.6 use science to extract value from our natural resources

3.7 participate in the technological development of Zimbabwe and the global world

4.0 OBJECTIVES

Learners will be able to:

- 4.1 apply scientific principles in solving everyday problems and in understanding new situations

- 4.2 conduct experiments, record results, describe observations and draw conclusions
- 4.3 demonstrate an understanding of scientific terms, laws, facts, concepts, theories and phenomena
- 4.4 exhibit knowledge and understanding in relation to scientific and technological applications with their social, economic and environmental implications
- 4.5 show relevant attitudes to science such as accuracy and precision, objectivity, integrity, enquiry initiative and inventiveness
- 4.6 portray knowledge and understanding of scientific instruments and apparatus including techniques of operations and aspects of safety.
- 4.7 use different forms of data presentation to give rational explanations of scientific phenomena.
- 4.8 draw scientific diagrams in two dimensions.
- 4.9 relate scientific principles, formulae and methods to solving qualitative and quantitative problems.
- 4.10 apply scientific principles, methods and techniques in value addition and beneficiation of our natural resources.
- 4.11 use appropriate methods of recycling and/or disposing wastes.
- 4.12 communicate scientific information logically and concisely

5.0 TOPICS

5.1 BIOLOGY:

- Laboratory rules and safety
- Cells and levels of organization
- Nutrition
- Respiratory systems
- Transport systems
- Reproduction in plants and animals
- Health and diseases

5.2 CHEMISTRY:

- Separation
- Matter

- Acids, Bases and Salts
- Oxidation and Reduction
- Industrial Processes
- Organic Chemistry

5.3 PHYSICS:

- Data presentation
- Measurements
- Force
- Energy
- Magnetism
- Electricity
- Robotics

6.0 METHODOLOGY AND TIME ALLOCATION

6.1 METHODOLOGY

Emphasis should be placed on providing learners with practical experience so that they see Science as an active and exciting learning area. Principles of individualisation, concreteness, totality and wholeness, self-activity and stimulation should under pin the implementation of teaching/learning methods in this learning area. The following methods are suggested:

- 6.1.1 Experimentation
- 6.1.2 Demonstrations
- 6.1.3 Problem solving
- 6.1.4 Field trips
- 6.1.5 Games
- 6.1.6 Cooperative learning/Group work
- 6.1.7 Simulations
- 6.1.8 Research/ Surveys/Case studies
- 6.1.9 Question and Answer
- 6.1.10 Discussions
- 6.1.11 Interviews and Report writing

6.1.12 Concept mapping

6.1.13 Visual tactile

6.1.14 Individualisation

6.2 TIME ALLOCATION

- For adequate coverage of the syllabus, a time allocation of 8 periods of 35 minutes per week is recommended. Double periods are recommended to accommodate practical work. The class size should not exceed 35 learners. At least 2 educational tours per year are recommended.

7.0 SCOPE AND SEQUENCE

7.1 BIOLOGY

TOPIC	Form 1	Form 2	Form 3	Form 4
Laboratory Safety Apparatus	- Laboratory rules and safety - Introduction to laboratory apparatus			
Cells and levels of organisation	- Plant and animal cell structure - Similarities and differences of plant and animal cells	Types of variation: continuous and discontinuous	- Structures and functions of specialised cells - Use of a microscope	Ecosystems
Nutrition	Diet	- Photosynthesis - Digestive system in humans	- Factors affecting rate of photosynthesis - Conditions necessary for photosynthesis - Teeth and digestion	- Human diet: - balanced diet - Deficiency diseases - Food tests
Respiratory system	Respiratory gases	- Respiratory organs - Breathing mechanism	Gaseous exchange in alveoli	Respiration: aerobic and anaerobic respiration

TOPIC	Form 1	Form 2	Form 3	Form 4
Transport systems	• Osmosis and diffusion • Components of blood	• Root and stem structure • Water and ion uptake • Heart structure and associated blood vessels	• Transpiration: factors affecting rate of transpiration • Measurement of transpiration • Plasmolysis • Turgidity • Blood circulation	• Adaptations of plants to reduce transpiration • Structure of blood vessels
REPRODUCTION IN PLANTS AND ANIMALS	• Reproduction in plants: flower structure, pollination, fertilisation, seed dispersal • Human reproductive organs • Puberty	• Detailed structure of a wind- and insect-pollinated flower • Functions of human reproductive organs	• Structure of the seed • Germination • Male and female reproductive systems • Sex cells • Fertilisation, pregnancy, placenta and child care • Menstrual cycle	• Asexual and sexual reproduction in plants • Inheritance • Methods of contraception • Contraceptives
HEALTH AND DISEASES	• Characteristics of a Health person • Importance of personal hygiene. • Waste disposal methods • Transmission of diseases	• Causes of diseases • Causes of bilharzia • Life cycle of bilharzia parasite • symptoms of bilharzia • Treatment of bilharzia • Prevention of bilharzia	• Sexually transmitted infections • Causes, signs and symptoms of gonorrhoea, syphilis, chancroid, and genital herpes • Control methods and treatment of STIs • Signs and symptoms of malaria, typhoid, Ebola and cholera	• HIV/AIDS • Immunity • Life cycle of malaria parasite • Methods of controlling malaria

TOPIC	Form 1	Form 2	Form 3	Form 4
			• Causes of malaria, typhoid, Ebola, cholera • Treatment of malaria, typhoid, Ebola and cholera • Effects of tobacco smoking on health • Effects of excessive consumption of alcohol • Effects of mandrax and cannabis • Effects of breathing solvents	

CHEMISTRY

TOPIC	Form 1	Form 2	Form 3	Form 4
SEPARATION	• Filtration • Magnetism • Winnowing • Decanting • Evaporation	• Applications of: filtration, magnetism, winnowing, decanting, evaporation	• Distillation, fractional distillation	• Paper chromatography

MATTER	• States of matter • Kinetic theory • Periodic Table: identification of elements • Metals and non-metals • Elements, mixtures and compounds	• Concentrations of solutions • Groups and periods	• Structure of atoms • Electronic configurations • Periodic table • Nuclide notation • Names and groups of elements • Mole concept, empirical and molecular formulae • Concentration	• Reactivity series • Factors affecting rate of reaction
• ACIDS, BASES AND SALTS	• Identification of acids and bases: litmus paper test	• Acid- base reactions	• pH scale • preparation of salts • reactions of metal + acid, acid + base, acid + carbonate	• Titration of bases with acid

INDUSTRIAL PROCESSES	• Production of peanut butter	• Production of soap	• Fractional distillation of liquid • Electrolysis • Electroplating	• Haber process • Contact process
OXIDATION AND REDUCTION	• Rusting: factors	• Chemical reactions: combustion	• Extraction of iron	• Alloy formation

PHYSICS

TOPIC	Form 1	Form 2	Form 3	Form 4
DATA PRESENTATION	• Tallies, tables, bar graphs	● straight line graphs	• pie charts, line graphs, interpretation	• pie charts, line graphs, interpretation and analysis

MEASUREMENT	- Physical quantities - S.I units	- Prefixes - Conversions - Density	- Physical quantities, accuracy and precision - Density of liquids	Derived quantities
FORCE	- Effects of force - Types of force - Measurement of force	- Moments of force - Resultant force - Levers	Weight/mass machines: inclined plane, pulleys, levers and gears	- Principles of pressure - Fluid systems - pumps
ENERGY	- Types of energy - Energy conversions	- Law of conservation of energy - Calculations on work - Properties of light - Sound energy	- Petrol and diesel engines - Heat transfer: convection, conduction and radiation	- Heat transfer applications: solar water heater and solar cooker - Telecommunications
MAGNETISM AND ELECTRICITY	- Magnets, poles, fields, compass - Static electricity: charges	- Properties of magnets - Attraction and repulsion	- Electromagnetism - Motor effect - Generator effect - Electroscope - Lightning	- Power generation - Electrical safety - Domestic electricity

	• Current electricity: conductors and insulators • Circuit components	• Circuit symbols and diagrams • Measurement of electricity	• Ohm's Law • Resistance • Resistors • Electrical power and energy	
• ROBOTICS • Principles of robotic systems	• Introduction to robotics -History of robots -Types of robots -Basic components of a robot	• Safety and ethics in robotics	• Sensors and actuators	
• robot design methodologies		• Tools and applications for robot design	• Logic gates • Tools and applications for robot design	
robot construction and programming		• Robot construction • Robot programming		• Robot construction - Robot programming

8.0 COMPETENCY MATRIX**FORM 1****8.1.1 BIOLOGY**

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
LABORATORY SAFETY AND APPARATUS	• explain laboratory rules • identify laboratory apparatus and their uses • demonstrate use of laboratory apparatus	• Laboratory rules • laboratory apparatus: beaker, measuring cylinder, test tube, crucible, evaporation dish, tripod stand, wire gauze, Bunsen burner, spirit burner, spatula, funnel	• discussing laboratory rules • discussing and drawing laboratory apparatus • taking readings from laboratory apparatus such as measuring cylinder, balances	• Multimedia & Braille/Jaws software • laboratory apparatus

CELLS AND LEVELS OF ORGANISATION	- describe the structure of a plant and an animal cell - list similarities and differences between plant and animal cells	- Plant cell: cell membrane, cell wall, cytoplasm, nucleus, permanent vacuole - Animal cell: cell membrane, cytoplasm, nucleus	Using bio viewers and hand lenses to observe and draw cells (avoid using fresh human blood or cheek cells)	- Charts on cells - Models of cells - ICT tools and braille/jaws softwares - Onion - Hand lens
NUTRITION	- define balanced diet - list component of balanced diet - name functions of nutrients	- Carbohydrates, proteins, fats, vitamins, fibre, mineral salts and water - Functions of nutrients	- Naming different kinds of foods and identifying their nutrients - Discussing functions of nutrients	- Charts on collection of foods - Food samples - E-learning packages
RESPIRATORY SYSTEM	- State the percentage composition of air - Identify respiratory gases - describe the test for carbon dioxide gas - test for oxygen gas	- Nitrogen 78%, oxygen 20%, carbon dioxide 0,03%, rare gases, water vapour less in air - Oxygen and carbon dioxide - Carbon dioxide	- Discussing the composition of air - Comparing oxygen in inhaled and exhaled air experiment - Blowing into lime water or bicarbonate indicator	- Model of breathing system - Internet - Inverted jar, candle, water trough - Lime water/bicarbonate indicator

		• Oxygen	• Using glowing wooden splint for oxygen test	• Glowing wooden splint
TRANSPORT SYSTEMS	• describe water movement in plants • identify components of blood stating the functions of each component	• Osmosis • Diffusion • Components of blood • Functions of the components	• Demonstrating water movement in a plant using potassium permanganate/ methylene blue/coloured solutions • Observing components of blood on slides	• use of potted plants • ICT tools & braille/ jaws software • Bio viewer, bioset
REPRODUCTION IN PLANTS AND HUMANS	• describe structure of a simple flower • describe pollination and fertilization • state signs of puberty	• Flowers, carpel, stamens • Pollen grains • Ovule, ovary, fruit • Signs of puberty, such as breasts, beard, menstruation, growth of pubic hair, widening of pelvic girdle, voice deepening	• Observing a flower using hand lens/ bio viewers • Emphasising on pre-menstrual symptoms such as period pain	• Charts on flowers • Flowers • Models of human being • Computer • Simulations • ICT tools
HEALTH AND DISEASES	• describe a healthy person	• State of being mentally, socially and physically well	• Discussing the state of a health person	• ICT tools • Disinfectants • Brooms

	• state importance of maintaining personal hygiene • list ways of disposing litter/ waste, stating their advantages and disadvantages • describe methods of transmission of diseases	• Importance of personal hygiene and food hygiene • Burying, recycling, burning • water, food, vectors, contact cholera – water ebola - contact malaria - vector bilharzia - vector	• Cleaning of classrooms • Carrying simple disinfection of drains • Discussing cleaning of a toilet • Picking and burying wastes • Discussing methods of disease transmission	• Mops • EMA resource person • Print media
--	--	---	---	--

8.1.2 CHEMISTRY

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
SEPARATION	● State methods of separating mixtures	• Filtration, Magnetism, Winnowing, Decanting, Evaporation	• Carrying out experiments to illustrate methods of separating	• Magnet • Sulphur • ICT tools
MATTER	• identify the three states of matter • describe properties of solids, liquids and gases in terms of kinetic theory of matters • describe the arrangement of particles in solids, liquids and gases	• solids, liquids and gases • properties of solids, liquids and gases • particle arrangement in solids, liquids and gases • Kinetic theory Elements, mixtures, Compounds	• Identifying the three states using salt, water and air • Compressing a gas in syringe • Discussing particle arrangement in solids, liquids and gases • Carrying out experiments on heating ice block, candle wax, naphthalene, iodine crystals	• salt, water, stones, syringe, beakers, ice, candles, naphthalene flakes • iodine crystals • models

	- identify mixtures, elements and compounds - Describe the factors that affect solubility - Identify metals and non-metals on periodic tables	Particle size, temperature, stirring Elements, mixtures and compounds Common elements of the periodic table	- Dissolving potassium permanganate crystals in water - Identifying elements on the Periodic Table - Mixing of iron and Sulphur, sand and maize grains - Heating iron and sulphur - Dissolving salt/sugar, iron/Sulphur in water - Identifying elements on periodic table	- Periodic Tables - iron filings - Sulphur powder - sand, maize - beakers - spatulas - salt/sugar - water
ACIDS AND BASES	- identify acids and bases using red and blue litmus - list properties of acids and bases	Acids and bases	Dipping litmus paper in HCL, NaOH, H₂O, CuSO₄, tap water	- Red/blue litmus - Solutions (acids and base)

INDUSTRIAL PROCESSES	- outline production of peanut butter - Outline the production of oil from peanut butter - State uses of oil	- Processes: Shelling, roasting, grinding and packaging - Equipment: sheller, peanut butter making machine	- Preparing peanut butter - Pressing peanut butter to produce oil	- Winnowing basket - Mortar and pestle - Grinding stone (guyo/imbokodo) - Roaster, sheller and peanut butter making machines
OXIDATION AND REDUCTION	- state conditions necessary for rusting - explain methods of preventing rusting	- Conditions: oxygen and moisture - Methods: painting, galvanising, plating	carrying out experiments to investigate conditions necessary for rusting	- iron nails - oil - multimedia - fused calcium chloride
ORGANIC CHEMISTRY	- identify forms of fuels - compare the efficiency of different fuels	Fuels: liquid, solid, gas	- Discussing forms of fuels - Carrying out experiments to compare the efficiency of fuels	- Wood, - Paraffin, - Methylated spirit burner, - Bunsen burners

8.1.3 PHYSICS

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
DATA PRESENTATION	- Present data in the form of tallies, tables and bar graphs - Interpret data presented in the form of tallies, tables and bar graph	tallies, tables and bar graphs	collecting and presenting data in the form of tallies, tables and bar graphs	multimedia
MEASUREMENTS	- estimate physical quantities - identify appropriate instruments for measuring physical quantities - measure accurately - identify types of errors in measurement - read an instrument scale to the nearest division	- Length, mass, time and temperature - SI units: metre (m), kilogram(kg), Kelvin(K) second(s) - Parallax error and zero error	- Estimating length, time, mass and temperature - Measuring length, time, temperature and mass - Taking measurements at different points	- Metre rule - Thermometer - Balance - Stop watch - ICT tools

	identify units including S.I units			
FORCE	- demonstrate the effect of forces on position, shape and size - identify various types of force - state the unit of force - identify instruments for measuring force	- Effects of Forces: deformation of solids, change of position, change in speed and direction - Types of forces: gravitational force, weight, mechanical force - electrostatic force, magnetic force and friction - S.I unit: newton (N) - Force meter - Spring balance	- Carrying out experiments on effects of forces - Carrying out experiments on gravitational force, push and pull, electrostatic force, magnetic force, friction - Measuring force using spring balance or force meter	- Foam rubber, springs, trolleys, rubber bands - Magnets, rulers, bricks - Trolleys - Masses - Force meter - Spring balance
ENERGY	- describe effects of energy - identify different -forms of energy	- Effects of energy - Forms and sources of energy	Demonstrating the effects of energy: burning fuels, bouncing a ball, making objects	spring, candle, ball, catapult, magnesium ribbon

	list forms of potential energy	- Kinetic, potential, light, heat, electrical, chemical, sound - Forms of potential energy: gravitational, elastic, chemical	- move, compressing a spring - Running upstairs, lighting a torch, clapping hands, raising a brick - Carrying out experiments to show potential energy	- torch, brick, musical instruments - trolleys - catapult - torch cell
	- identify energy conversions - construct energy chains - identify energy convertors	- Energy conversion: energy changes, energy chains - Energy convertors	- Lighting a torch - Using a dynamo to light a bulb - Using a catapult and solar panel - Discussing various energy convertors	- Torch - Dynamo - Catapult - Solar panel - Bulb - Green plants - Engines - refrigerator - Cell - Bulb - Solar cells - Generators
MAGNETISM	identify types of magnets	Magnets: bar magnets, horse	Classifying magnets	Horse shoe magnets

	- distinguish magnetic material from non-magnetic materials - identify the poles of a magnet	shoe magnets, C-magnets, E-magnets - Magnetic and non-magnetic materials - Poles and magnets: South and north poles: earth as magnet	- Using magnets to identify magnetic and non-magnetic materials - suspending a bar magnet to determine polarity - using a compass	- Bar magnets - C-magnets - E-magnets - Steel, iron nails, copper, aluminium, wood, plastic, glass - bar magnet, string, compass,
ELECTRICITY	- state the two types of charges - describe the production of charges - define current - distinguish between conductors and insulators	- Negative, positive - Static electricity - Current as flow of charges - Conductors and insulators	- Rubbing polythene or perspex on hair or dry cloth - Carrying experiment out on	- Polythene - Perspex - Dry cloth - Plastic - Wood - Iron bar electrolytes, - Carbon rods,

	• identify components of direct current (d.c) circuit • draw and label a simple direct current (d.c) circuit	• Cells, batteries, bulbs, switches, resistors, standard symbols	conductivity of different materials • Drawing diagrams circuit using symbols	• Copper rods • Light bulb • Voltmeter • Ammeter • Circuit boards • Cells • Bulbs • Switches • Resistors • Chart of standard electrical symbols • ICT tools
ROBOTICS Principles of robotic systems	• Define robotics and robot • Outline the history and evolution of robotics • identify types of robots • List basic components of robots	• Evolution of robotics • Fixed and mobile robots • Basic components of a robot - motion sensors - gears - Micro controller such as ESP32,	• Discussing the history and evolution of robotics • Identifying types of robots • Naming components of robots	• ICT tools • Robotic kits • Charts • Camera • LED • LCD • Memory card • Drones

		ESP8266 and STM32 - power source -Motors (actuators) such as Savor motor, Stepper motor, DC motor and Vibration motor - motor drive		
--	--	--	--	--

FORM 2

8.2.1 BIOLOGY

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
TYPES OF VARIATION	- state differences among living organisms - compare continuous and discontinuous variation	variation: height, mass, shoe size, skin/ coat colour, sex, right or left handedness, tongue rolling	comparing height, mass, shoe size, skin/ coat colour, sex, right or left	- leaves - seeds in a pod - ICT tools

	draw bar graphs to show variations in living organisms	bar graphs	handedness, tongue rolling draw bar graphs showing variations	
NUTRITION	- explain importance of plants as producers - state the word equation for photosynthesis - draw and label digestive system of humans - outline the route followed by food in the human digestive system	- Animals as consumers of food - Photosynthesis - Digestive system in Humans: alimentary canal	- Discussing the conversion of solar energy to chemical energy by plants - Observing a model of the alimentary canal	- Charts on leaf structure - ICT tools - Models of the alimentary canal - Charts on human alimentary canal
RESPIRATORY ORGANS	- state word equations for respiration - label parts of respiratory system	$\text{Glucose} + \text{Oxygen} \rightarrow \text{carbon dioxide} + \text{water}$	Carrying out experiments to show that energy is released during burning of food (during respiration)	- Models of respiration - ICT tools

TRANSPORT SYSTEMS	- Outline the internal structures of a root and stem - describe water and ion uptake by plants - draw and label the structure of the heart - describe the circulatory system - name the main blood vessels to and from the heart - state functions of the heart	- Root and stem structures of a dicotyledonous plant: epidermis, cortex, phloem, cambium, xylem - Osmosis and active uptake - Human circulatory system - structure of the heart, atria, ventricles and valves; bicuspid and tricuspid, semi-lunar - receiving and pumping blood - Blood vessels and heart - Movement of blood	- Discussing the internal structures of a root and stem - Viewing of prepared slides - Carrying out an experiment to show arrangement of vascular tissue using dye - Examining model of the heart - Drawing and labelling structure of the heart - Identify vessels to and from the heart on a model - Simulations on action of the heart	- Plant - ICT tools - Microscope - Bio-viewer - Model of human circulatory system - ICT tools
--------------------------	--	---	---	--

REPRODUCTIVE SYSTEMS	- distinguish between monocotyledonous and dicotyledonous plant seeds - describe functions of cotyledon and endosperm - state the functions of the female and male reproductive systems	- Structure of maize and bean seeds: testa, radicle, plumule, endosperm and cotyledon - Human reproductive systems: Penis, testis, sperm duct, urethra, prostate glands, vagina, ovary, oviduct, uterus, cervix	- comparing internal and external features of the monocotyledonous and dicotyledonous plant seeds - discussing the functions of male and female reproductive systems	- multimedia - models of reproductive systems
HEALTH AND DISEASES	- state causes of diseases - describe causes of bilharzia - describe the life cycle of bilharzia parasite - identify the signs symptoms of bilharzia - describe the treatment of bilharzia	- Contaminated food, water, air - Bilharzia worm - Snail-human host - Eggs in water - Urine with blood and faeces stained with blood, fever, fatigue - Anti parasitic drugs - Avoid playing stagnant water	- discussing the transmission of bilharzia parasite - describing the bilharzia parasite life cycle - identifying the signs and symptoms of the bilharzia parasite	- bilharzia life cycle chart - multimedia

	Outline methods of preventing the spread of bilharzia		- describing the treatment of bilharzia - discussing methods of preventing spread of bilharzia	
--	---	--	---	--

8.2.2 CHEMISTRY

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
SEPARATION	state the applications of filtration, winnowing and magnetism and evaporation	- application: treatment of water, grain separation, separation of metallic objects from grain before grinding, separation of metallic waste for recycling - formation of sugar and	discussing applications of the separation methods	- multimedia - field trips

		production of ammonium nitrate crystals from their solution		
MATTER	- determine the concentrations by colour intensities of dissolved substances - determine the concentration of a substance by varying the amount of solute in a given solvent	- Mole concept: - concentration of solution expressed in terms of mass of solute dissolved per given volume of solvent	- carrying out experiments on dissolving-coloured substances (differing amounts to be dissolved) - carrying out experiments on dissolving known masses of solute in known volumes of solvent	- potassium permanganate - ammonium dichromate - copper sulphate - brown sugar - salt, water - ammonium nitrate
ACIDS, BASES AND SALTS	describe an acid–base reaction	Acids–base reactions	Carrying out experiments to demonstrate acid- base reactions	- litmus paper - Dilute HCl, H₂SO₄ and NaOH solution
INDUSTRIAL PROCESSES	Outline the manufacture of soap	Production of soap/ saponification	- Making soap - Visiting soap manufacturing companies	NaOH solution, NaCl solution, plant /animal fat

OXIDATION AND REDUCTION	• write simple word equations • define oxidation and reduction in terms of oxygen • distinguish between physical and chemical changes	• Chemical reactions	• burning Magnesium ribbon • melting ice, burning sugar/mealie- meal	• burner • Magnesium ribbon and mealie meal/sugar
ORGANIC CHEMISTRY	• define complete and incomplete combustion of fuels • list the products of complete and incomplete combustion of fuels • describe the effects of burning fuels	• Complete and incomplete combustion • Global warming, deforestation,	• lighting burners (methylated spirit) with long and short wicks/ paraffin stove • lighting Bunsen burner sleeve wide/narrow • Carrying out environmental awareness campaigns	• Burner (paraffin/ methylated) • Gas burners • Posters, drama.

8.2.3 PHYSICS

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
DATA PRESENTATION	- construct a straight-line graph from appropriate data - interpret straight line graphs	Straight line graphs	Drawing a straight-line graph from given data	multimedia
MEASUREMENTS	- convert units - measure mass of a liquid - measure the volume of an irregular object - determine the thickness, volume and mass of small objects	- prefixes of S.I units - Measurement of mass - Measurement of volume by displacement - Measurement of volume, thickness and mass of small objects	- Converting metre to centimetre, millimetre, kilogram to gram, minutes to second - Carrying out experiments on measurement by differences of mass - Carrying out experiments on measurement of irregular objects - Carrying out measurements of	- Metre rule, balance, stopwatch, - Beaker and water - Measuring cylinder - Irregular objects - Overflow can - String

FORCES	• calculate density • calculate the resultant of a pair of inline forces • define moment of a force • calculate moment of a force • state the principle of moments • apply the principle of moments in simple calculations • define friction • measure friction • state the applications of frictional force	• Formula for density • Balanced and unbalanced forces • Resultant force • Moment of force = force $\times$ perpendicular distance from the pivot • At equilibrium: Clockwise moments = anti clock wise moments • Frictional forces: forces opposing motion • nature of surface	volume, thickness and mass • carrying out experiments to find mass and volume • Carrying out experiments to demonstrate equal and unequal forces • Demonstrating moments of force • applying the principle of moments • Carrying out experiments to investigate frictional forces	• Book, seeds, pins • Metre rule • Balance • Stopwatch • Regular and irregular objects, liquids • force meters, levers • masses, balance, metre rule, • spring balance
----------------------	--	--	--	---

		• car braking system • tyre treads • soles of shoes • road surfaces		
MACHINES	• define a machine • construct a simple machine	• Simple machines: levers	• Lifting a load using a crowbar • Constructing a simple machine	• Crowbar • Planks • masses
ENERGY	• state the law of conservation of energy • define work and energy • state the S.I unit of work and energy • calculate the work done or energy used by forces • list sources of light energy • show that light travels in a straight line	• Conservation of energy • Work and energy • Joule (J) • Work done or energy used = force x distance moved • Sun, fire, electric bulb • Production of shadows	• Carrying out experiments to demonstrate Work done = Energy used • Solving problems on work done and energy used • observing sources of light • carrying out experiments to demonstrate that light	• force meter • object • ruler • bulb, torch, fire, candle, sun • light source, slits screen • ball • multimedia

			travels in a straight line (production of shadows)	
	- state the production of and transmission of sound - demonstrate the need for a medium in the transmission of sound.	Sound Energy - vibrations - passing of sound energy through different media	- Producing sound using musical instruments - Conducting a bell jar experiment to show that sound requires a material medium for transmission	- musical instruments - Tuning fork - bell jar - vacuum pump - electric bell
MAGNETISM AND ELECTRICITY	- describe properties of magnets - state the law of magnetism - draw magnetic fields	- Polarity, field direction and strength - Attraction, repulsion, like and unlike poles - Magnetic fields	- Illustrating properties of magnets - Carrying out experiments using magnets - Demonstrating magnetic fields	- bar magnets, string, compass - bar magnets - iron filings, magnets, plain paper

	- define current and voltage - state the S.I units of current and voltage - measure current and voltage - determine electrical power	- Current: ampere (A) Voltage: Volt (V) $\text{Power} = \text{Current} \times \text{Voltage}$	- Discussing the meaning of current and voltage - Carrying out experiments to measure current and voltage - Calculating electrical power	- Ammeter - Voltmeter - Circuit boards - Connecting wires - Light bulbs
Principles of robotic systems	Discuss safety and ethics in robotics	- Safety in robotics - Ethics in robotics	Discussing safety and ethics in robotics	- ICT tools - Robotic kits
Robot design methodologies	- Identify tools and applications for robot design - use tools and applications for robot design	CAD tools, simulation software, micro controllers (Arduino)	Computer simulation and coding (scratch)	Resource persons
Robot construction and programming	Construct and code	Robot design and construction	Constructing robots	

FORM 3**8.2.1 BIOLOGY**

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
STRUCTURE AND FUNCTION OF A CELL	• identify specialized cells • draw and label specialised cells • state the function(s) of the specialised cells in relation to structure • use a microscope to observe cell structure	• Specialised cells -red blood cell -muscle cell -palisade cell -root hair cell • Functions of specialised cells	• Drawing specialised cell structures • Using microscope/bio-viewer to observe specialized cells Use of a microscope only. (Identification of parts not necessary)	• Microscope • Charts • Bio-viewers • Bio-sets • Printed multimedia • ICT tools • Prepared slides

			(Avoid using fresh, animal cells)	
NUTRITION Factors affecting the rate of photosynthesis	- state the factors which affect photosynthesis - describe experiments on factors which affect the rate of photosynthesis - describe the fate of end products of photosynthesis - identify parts of the internal structure of a leaf - describe how the leaf is adapted for photosynthesis	- Water, light carbon dioxide, chlorophyll - Carbohydrates and oxygen - Translocation, storage and structure formation, respiration - Epidermis, stomata, vascular tissue, mesophyll, guard cells - Surface area, stomata, palisade cells, air spaces.	- Investigating factors affecting rate of photosynthesis - Testing for starch in a leaf - Testing for oxygen using a glowing splint - Observing the internal leaf structure using a bio-viewer/microscope - Examining the external leaf structure	- Iodine solution - Leaves - Glowing splint - Light source - Pond weed - Sodium hydrogen carbonate/Soda lime/Sodium hydroxide - Microscope - Bio-viewer - ICT tools

HUMAN NUTRITION Alimentary canal	- draw and label parts of the alimentary canal and associated organs - identify parts of the alimentary canal of a human - state the functions of parts of the alimentary canal of a human	- Mouth, oesophagus stomach, small and large intestines, gall bladder, pancreas, liver - Ingestion, digestion, absorption, assimilation and egestion	- Watching video simulation to examine human alimentary canal - Identifying parts using a chart	- ICT tools - Recommended textbooks - model of human alimentary canal.
NUTRITION	- Name types of teeth and their functions - Describe mechanical and chemical digestion - Explain the importance of digestion - Describe the function of a typical enzyme (amylase)	- Incisor -cutting -biting - Canines -tearing -gripping - Premolar and molar -grinding Chewing - Mechanical digestion-breaking down into smaller pieces chemical – use of enzymes to	- Discussing the functions of teeth and their structure - Observing types of teeth on the skeleton models - Using a model to examine action of amylase on starch - discussing mechanical digestion	- ICT tools - Skeleton model - Visking tubing/egg shell - Recommended textbooks

	Identify the end products of digestion	convert from insoluble to soluble smaller molecules - Increase surface area for enzyme action and solubility - Catalysis for conversion of starch to maltose/glucose - carbohydrates: glucose - Protein: amino acids - fats: fatty acids and glycerol	- demonstrating increase in surface area when solid is broken into smaller pieces. - experimenting using enzymes on different food substrates. - testing for glucose, proteins and fats.	
RESPIRATORY SYSTEMS Gaseous exchange in alveoli	- state the differences between inhaled and exhaled air - describe the role of the alveoli in gaseous exchange - explain how the alveoli is adapted for gaseous exchange	- Percentage composition of inhaled and exhaled air - Diffusion of carbon dioxide and oxygen - Adaptations: 1 cell thick, moist, large surface area, network of blood capillaries	- Carrying out an experiment to show the change in proportions of carbon dioxide and oxygen in inhaled and exhaled air - Describing gaseous exchange in the alveoli	- ICT tools - Limewater/bicarbonate indicator solution - model of alveoli

			discussing adaptations	
TRANSPORT SYSTEMS Transpiration	- explain the process of transpiration - state factors affecting the rate of transpiration - measure transpiration in a plant - outline the importance of transpiration	- Water loss in plants - Wind speed, temperature, humidity, surface area, light intensity, number of stomata - Use of a potometer - Water and mineral salts uptake, cooling the plant	- Discussing the process of transpiration - Carrying out experiments to investigate factors affecting the rate of transpiration - Explaining the importance of transpiration	- ICT tools - Potometer
Plasmolysis and turgidity	- explain the terms plasmolysis and turgidity - describe the effects of water loss and water gain in plant cells	- Plasmolysis - Turgidity	- Discussing plasmolysis and turgidity - Watching video simulations on plasmolysis and turgidity - carrying out experiments to demonstrate plasmolysis and	- ICT tools - Potato strips - Visking tubing

			turgidity using potato strips	
Blood circulation	describe the double circulatory system	human blood circulatory system	- Watching video simulations on the blood circulatory system - examining the model of the human circulatory system.	- Multimedia - Circulatory System Model
REPRODUCTIVE SYSTEMS Wind and insect pollinated flowers	explain the structures of wind and insect pollinated flowers	- Wind flower pollinated - Insect flower pollinated	- Examining the structure of wind and insect pollinated flower - Drawing the structures of wind and insect pollinated flower	- Plant specimen - Multimedia
Germination	- describe the process of germination - investigate conditions necessary for germination	- Germination - Moisture, warmth, oxygen	- Discussing the process of germination - Carrying out experiments to demonstrate conditions	- Seeds - Multi-media

	calculate percentage germination	Percentage germination	- necessary for germination - Determining germination percentage	
Reproductive systems	- state the functions of the male and female reproductive system - describe the structure and functions of sex cells - draw and label the sex cells - describe the route of sperm from the testes to the ovum after ovulation - define fertilisation	- male and female reproductive system - functions - Sperm and ovum - epididymis, testis, sperm duct, urethra, vagina, cervix, uterus, oviduct	- drawing the male and female reproductive systems - Drawing the structure of sex cells	- models of reproductive organs - ICT tools - Multimedia - Bio-viewers
	describe the menstrual cycle	- Hormones i.e. progesterone, oestrogen - Ovulation - Menstruation	Illustrating the menstrual cycle	Multi-media

	- describe the role of the placenta - identify the substances exchanged in the placenta	- Placenta - Nutrients, wastes, antibodies, oxygen	- Watching video simulations of the placenta - Listing the substances exchanged in the placenta	Multi-media
HEALTH AND DISEASES	- list sexually transmitted infections (STIs) - describe the signs, symptoms and effects of: Gonorrhoea, Syphilis, Chancroid, Genital herpes - state the causative agents of:	- Gonorrhoea - Syphilis - Chancroid - Genital herpes - Pathogens -virus -bacteria	- Discussing STIs - Identifying the sign, symptoms and effects of STIs - Watching videos on signs and symptoms of STIs - Listing the causes	- Bio-viewers - Multi-media

	- Gonorrhoea, Syphilis, Chancroid, Genital herpes - state the control methods and treatment	Abstinence, condoms, contact tracing		
	- describe the signs and symptoms of malaria, typhoid, ebola and cholera - state the causes of malaria, typhoid, ebola and cholera - explain how the diseases are treated	- Malaria - Typhoid - Ebola - Cholera	- Using the signs and symptoms - Discussing the causes - Outlining the disease control methods	Print media
	- describe the effects of tobacco smoking on health - describe effects of excessive	- Emphysema, bronchitis, lung cancer, low birth weight - Liver cirrhosis, social implications,	- Discussing the effects of smoking - Identifying effects of excessive alcohol consumption	Multi-media

	consumption of alcohol • explain effects of use of mandrax and cannabis • outline the effects of breathing solvents	reduced reaction time • Hallucinations, addiction • Damaged muscles, heart and addiction	• Listing effects of mandrax and cannabis • Explaining effects of breathing solutions	
--	--	---	--	--

8.2.2 PHYSICS

TOPIC	OBJECTIVES Learners should be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
DATA PRESENTATION	• construct a pie chart • Interpret and analyse data from pie charts and line graphs	• pie charts and line graphs	• explaining data presented in form of pie chart and line graphs	• multimedia
MEASUREMENT Physical quantity	• measure physical quantities accurately using appropriate instruments	• Measurement of physical quantities: length (thickness and	• Measuring length, current, voltage	• Vernier callipers • Voltmeter • Ammeter

	• read instruments scale to the nearest fraction of the division • determine density of liquids	internal diameter) current, voltage • Density: liquids	• Determining density of liquids experimentally	• Measuring cylinder • Balance • Multimedia
FORCE Weight/mass	• define weight, momentum and inertia • distinguish between weight and mass • state Newton's laws of motion • calculate force, given mass and acceleration • state the application of Newton's laws of motion	• Weight, • Momentum • Inertia • $\text{force} = \text{mass} \times \text{acceleration} (F=ma)$	• Defining weight, momentum and inertia • Discussing Newton's laws of motion limited to linear motion • Verifying Newton's second law of motion experimentally	• Spring balances • Trolleys • Ticker timer

Machines	- describe the uses and applications of machines - determine Mechanical Advantage (MA), Velocity Ratio (VR) and efficiency of levers inclined plane, pulleys and gears - explain energy losses in machines - describe ways of improving efficiency in machines	- Simple machines: Levers, pulley system, inclined plane, gears - Mechanical advantage, velocity ratio and efficiency - Friction and mass of moving parts - Lubrication, bearings and mass reduction	- Lifting different loads using the simple machines - Carrying out experiments to determine MA, VR and efficiency - Calculating mechanical advantage and velocity ratio of the simple machines - Demonstrating effects of lubrication, bearings and mass reduction on efficiency	- Crow bar - Wheel barrow - Scissors - Pulleys - Inclined plane - Gears - Bearings
Petrol and diesel engines	- describe the operation of a four-stroke petrol and diesel engine - explain the role of the fuel injector and carburettor	- Strokes: intake compression, power and exhaust - Fuel and air supply - Computer box	- Demonstrating intake, compression, power and exhaust using an engine model - Explaining how the fuel injector and the	- Model of a four-stroke petrol engine - Engine block - model of a modern petrol engine

	- outline the difference between petrol and diesel engines - describe the operation of modern petrol and diesel engine. - outline the advantages of modern petrol engines over old petrol engines	- Ignition methods, efficiency, carbon monoxide production - modern petrol engine - fast start, efficiency	carburettor works and their importance - Examining the ignition and fuel economy of petrol and diesel engine - field tour to a modern garage	- model of a modern diesel engine - car engines - Fuel injector - Modern garage
ENERGY heat transfer	- Give an account of the heat transfer - explain convection in terms of the kinetic theory of matter - state good and bad reflectors, absorbers and emitters of heat	- conduction, good and bad conductors - convection in liquids and gases - good and bad reflectors, absorbers and emitters of heat	- carrying out experiments with metal and non-metal rods - demonstrating heat movement in liquids and gases - carrying out experiments on absorption, emission and reflection of heat using different surfaces	- iron - copper - aluminium - wood - glass - smoke chamber - copper sulphate crystals - black painted surface

				white painted and shiny surfaces
MAGNETISM Electromagnetism	describe an experiment to demonstrate that a current carrying conductor has a magnetic field around it.	Magnetic field patterns of long straight conductor, solenoid	Demonstrating magnetic field around a current carrying conductor, using iron filings and plotting compass	- Copper wire - Iron filings - Source of direct current - magnetic compass
Motor effect	Illustrate how movement is produced in a current carrying wire in a magnetic field	Motor effect: Interaction of magnetic fields, electrical energy converted to kinetic energy, motion	Carrying out an experiment using a current carrying wire between magnets	- Copper coil, source of direct current - Magnets
	- describe the operation of a direct current (d.c) motor - state the factors that affect the rotation of the coil	- electrical to mechanical energy - strength of the magnetic field - number of turns on the coil - amount of current passed	constructing and operating an electric motor	- electric motor - model of a motor

Generator effect	describe an experiment which shows that a change in magnetic field can induce an electro motive force (e.m.f) in a conductor	the generator principle, magnetic field, electric field and e.m.f	carrying out an experiment which illustrates the generator effect	- copper coil - galvanometer - magnet - source of d.c
	- state the factors which affect the magnitude of - induced e.m.f	strength of a magnet, relative motion, number of turns and area of coil	carrying out experiments to investigate factors which affect the magnitude of induced e.m.f.	- magnets of different strengths - coils of different turns - coils of different area
	describe the operation of an alternating current (a.c) and d.c generator	mechanical energy to electrical energy	- illustrating output voltage graphically - discussing factors affecting output voltage	- a.c generator - d.c generator - multimedia
STATIC ELECTRICITY Electroscope	Describe simple experiments to show electrostatic charging using an electroscope	Electron, positive and negative charges	- Demonstrating electrostatic charging - Demonstrating attraction and repulsion of charges	- Electroscope - gold leaf, aluminium foil - nail

	Describe forces between charges	like and unlike charges	using an electroscope	
Lightning conductor	- describe the production of lightning - Explain the principle of a lightning conductor - state the dangers of lightning - state safety precautions against lightning	- Movement of charges between cloud and ground - Height, conduction and earthing - High voltage electrocution heating effect	- Discussing the production of lightning - Erecting a lightning conductor - Discussing dangers of lightning - demystify myths on lightning-wearing red clothes	Multimedia
Current Electricity Ohm's law	- state Ohm's law - calculate resistance /voltage/current using Ohm's law - describe an experiment to determine resistance - state the limitations of Ohm's law	- Ohm's law $V=IR$	- Carrying out experiments to verify Ohm's law - Carrying out calculations using Ohm's law	- Voltmeter - Ammeter - Bulbs - Resistors - Different conducting materials

	- state the factors that - affect resistance	- resistance - limitations of Ohm's law - factors affecting resistance: length of wire, thickness of wire (cross sectional area)	- determining resistance experimentally using a voltmeter and an ammeter - discussing limitations of Ohm's law	
Resistors	- construct simple electric circuits - draw and interpret circuit diagrams - calculate resistance in simple circuits	- simple electric circuits: cells, switches, resistors, variable resistors, bulbs ammeter, voltmeter and fuses - series and parallel arrangement of resistors	- setting up practical electric circuits - drawing and interpreting circuit diagrams - determining total resistance of resistors in series	- cells - resistors - switches - bulbs - fuses - ammeters - voltmeters

		• $R_{\text{total}} = R_1 + R_2 + R_3$ • $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$	and parallel arrangement	• leads • Resistors connected in series and parallel
Electrical power and energy	• Define electrical power and energy • Calculate electrical power and energy	• Power and energy • Units: • watt (W), kilowatt-hour (kWh) • $P = VI$ • $E = VIt$	• Discussing electrical power and energy use by appliances • Determining power and energy using the formulae	• Electric iron • Electric fan • Heating elements • Smartphones • computers • multimedia

STATIC ELECTRICITY Electroscope	- Describe simple experiments to show electrostatic charging using an electroscope - Describe forces between charges	- Electron, positive and negative charges - like and unlike charges	- Demonstrating electrostatic charging - Demonstrating attraction and repulsion of charges using an electroscope	- Electroscope - gold leaf, aluminium foil - nail
---	---	--	---	---

Principles of robotic systems Sensors and actuators	• identify sensors and actuators • explain the functions of the sensors • describe function of actuators	• Temperature sensors, ultrasonic sensors, light sensor, pressure sensor, proximity sensor, • Actuators - electric, -hydraulic - pneumatic	Identifying sensors and actuators • Discussing functions of sensors and actuators	• online resources • ICT tools • Robotic kits • Resource persons • Braille/jaws software
---	--	--	--	--

Robot design methodologies	• identify tools and applications for robot design • apply tools and applications for robot design	• CAD tools, simulation software, micro controllers (Arduino) •	• Computer simulation and coding (scratch) • Writing structured programs to control robots for defined tasks	• ICT tools • Robotic kits • Resource persons • Braille
-----------------------------------	---	--	---	--

Robot construction and programming Logic gates	• Describe logic gates • Construct and code •	• Logic gates: NOT, OR, AND, NAND, NOR and Truth tables • Robot design and construction • Coding – Python programming •	• Describing logic gates • Constructing robots •	• ICT tools • Robotic kits • Resource persons • Braille
--	---	--	--	--

8.2.2 CHEMISTRY

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
SEPARATION	describe the processes of distillation and fractional distillation	distillation and fractional distillation	demonstrating distillation of impure water and fractional distillation of dilute ethanol	- ethanol - distillation unit - fractional distillation unit - charts

MATTER	- describe the arrangement of metals and non-metals in the Periodic Table - name the sub-atomic particles - state the relative charges and masses of sub-atomic particles - state relative position of sub-atomic particles within the atom - name the first 20 elements in the periodic table stating their symbols - write the electronic configuration of the first 20 elements - describe ionic and covalent bonding	- metals and non-metals: -position - Proton, neutrons and electrons - Structure of an atom: nucleus, electron shells - Electronic configurations - Bonding - Ionic bonding: NaCl, MgO, Na₂O - Covalent bonding: H₂, Cl₂, H₂O	- Identifying positions of elements on the Periodic Table, their groups and periods - Drawing the structure of an atom - Producing electronic configuration diagrams - Burning magnesium ribbon to demonstrate chemical reaction - Experimenting to form precipitates: formation of lead chloride/silver chloride/barium sulphate	- Periodic Table chart - Computer simulation of atomic structure - Magnesium ribbon - Lead nitrate, HCO₃, H₂SO₄ - Barium nitrate - Silver nitrate
---------------	--	---	---	---

	- define relative mass/mass number - define the proton number/atomic number - calculate the number of neutrons from given data - define isotopes	- nuclide notation ${}_b^aX$ - Isotopes: ^{16}O and ^{18}O ^{35}Cl and ^{37}Cl ^{12}C and ^{14}C	- Explaining relative mass/mass number using models - Defining isotopes	Periodic Table chart
	- Define the Avogadro number - State the relationship between the mole and molecular mass (M_r) or atomic mass (A_r) - Calculate empirical formula and molecular mass - Calculate concentration of solutions in mole/dm^3 and g/dm^3	- Avogadro number - Mole concept $n = \frac{m}{M_r}$ - Empirical formula and molecular formula from percentage composition data - Concentration	- Stating Avogadro constant - Calculating molecular masses - Calculating empirical formula - Calculating concentration of solutions	Models/use of marble to demonstrate moles
ACIDS AND BASES	identify the regions of acidity neutrality and alkalinity of substance on the pH scale and using	- The pH scale - Use of universal indicator solution	- Drawing the pH scale - Carrying out experiments using	- pH scale chart - Universal indicator solution

	universal indicator solution - describe the reactions of acids with metal and bases - write word and chemical equations for reactions	- Reaction of dilute acids with metals, bases and carbonates - neutralization	universal indicator to identify the pH ranges for different substances Reacting dilute hydrochloric acid, nitric acid and sulphuric acid with magnesium, zinc, calcium carbonate, magnesium carbonate, sodium hydroxide	- Ammonia solution, sodium hydroxide, vinegar/lemon juice, hydrochloric acid, water - Dilute acids - Hydroxides - Metal granules/powder
INDUSTRIAL PROCESSES	- outline the production of nitrogen and oxygen - define electrolysis - label the general components of an electrolytic cell - list properties of electrodes and electrolyte - describe anode and cathode reactions for	- Fractional distillation of liquid air - Electrolytic cell: connecting wires, cathode, anode, battery, electrolyte - Electrolysis of: molten lead bromide - Solid lead and bromine fumes - Electrolysis of water	- Discussing the production of nitrogen and oxygen - Setting up an electrolytic cell - Setting out experiments on electrolysis of water - discussing electrolysis of molten lead bromide - experimenting on electrolysis of water	- Flow charts - Battery, electrodes molten lead bromide - Dilute H_2SO_4 - Carbon electrodes

	electrolysis of molten lead bromide - state observations for the electrolysis of molten lead bromide - describe the electrolysis of water - state the products formed during the electrolysis of water - state the uses of oxygen and hydrogen	- Electrolyte: dilute H_2SO_4 O_2 and H_2 H_2: Haber process O_2: basic oxygen furnace, medical purposes	discussing uses of oxygen and hydrogen	
	- state the cathode, anode and electrolyte - explain the cathode process - state reasons for electroplating materials	- Copper electroplating an iron nail - Cathode reaction - Decoration - Prevention of corrosion	- Setting out experiments to electroplate an iron nail - identify electroplated objects - Discussing reasons for electroplating in groups	- Iron sheet/nail - Copper sulphate solution - Copper electrode

OXIDATION AND REDUCTION	- define oxidation - define reduction - list the raw materials used in the extraction of iron and their sources	- Oxidation: loss of electrons, gain of oxygen, loss of hydrogen - Reduction: gain of electrons, loss of oxygen, gain of hydrogen - Redox reactions - Extraction of iron at ZISCO steel in the blast furnace - Lime stone/calcium carbonate/CaCO_3, iron ore/haematite/Fe_2O_3coke/carbon/C	- Demonstrating the reaction of copper oxide with hydrogen - Discussing the extraction process - Visiting ZISCO steel	- Copper oxide - Hydrogen gas - ZISCO steel
	- describe reactions in the blast furnace - state the functions of the raw materials - describe how iron and slag separate	Reactions in the blast furnace: formation of CO_2, formation CO reduction of Fe_2O_3, decomposition of CaCO_3, formation of slag	Heating sample of Iron (III) Oxide on a charcoal block	- Iron oxide - charcoal

ORGANIC CHEMISTRY	- define the term hydrocarbon - name the members of the homologous series with 3 carbon atoms - draw the displayed structures of methane, ethane, propane, ethene, propene - state uses of methane, ethane, propane, ethene, propene.	- Hydrocarbons: - alkanes - alkenes - methane - ethane - propane - ethene - propene - drawing structures - as fuels and alcohols	Outlining the structures of methane, ethane, propane, ethene and propene using models	Models of atoms and bonds
	- outline the production of biogas - identify factors affecting the production of biogas - state the use of biogas	- Biogas production: - role of bacteria - temperature - pH - Fuel	Preparing a model of a bio digester	- Model of bio digester - Samples of organic waste (plant and animal waste)

FORM 4**8.2.1 BIOLOGY**

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
CELLS AND LEVELS OF ORGANISATION Ecosystems	• explain the term ecosystem • list components of an ecosystem	• Organisms and their environment • Physical and biological components	• Discussing the term ecosystem • Identifying components of an ecosystem • Examining various ecosystems e.g. garden, pond, forest	• Multi-media
	• explain natural ecosystem • construct food chains, food webs and pyramids of biomass	• Natural ecosystem • Food chains, food webs and pyramids of biomass • Energy flow	• Engaging in field trips to observe natural ecosystem • Illustrating food chains, food	• Multimedia

	• explain how energy is lost in food chains and food webs • describe the carbon and nitrogen cycles • describe an artificial ecosystem • explain bio-diversity • identify problems caused by limited bio-diversity • state advantages of biodiversity	• Carbon and nitrogen cycles • Garden, pond and plantation • Bio-diversity • Soil infertility, pests and diseases • Wide variety of food source • Self-sustenance of an ecosystem • Interdependence • Less spread of diseases	• webs and pyramids of biomass • Discussing energy input and energy flow • Explaining the carbon and nitrogen cycles • Visiting plantations, pond, garden • Discussing bio-diversity • Explaining problems caused by limited bio-diversity	• Multimedia
--	--	--	---	--

NUTRITION	plan a balanced diet meal for different groups of people	Diet for toddler, adolescent, manual worker, sedentary worker	- Planning balanced meals - Discussing balanced diet	- Food samples - Multimedia
	- describe the functions of the components of a balanced diet in the body. - Explain malnutrition - explain the term deficiency disease - state causes of deficiency diseases	- Functions of carbohydrates, fats, proteins, vitamins (A, C, D), mineral salts (iodine, calcium, iron), water, roughage - Effects of malnutrition, obesity, anorexia nervosa - Deficiency diseases: - Kwashiorkor, goitre, rickets, scurvy, anaemia and night blindness	- Identifying components of a balanced diet - Discussing functions of the nutrients in the body - Discussing diabetes mellitus 2, obesity and anorexia nervosa - Explaining deficiency diseases - watching videos and pictures of people suffering from deficiency diseases	

	Describe the tests for starch, simple sugars, protein and fats	Food tests	- Listing causes of various deficiency diseases - Carrying out experiments to test for starch, simple sugars, protein and fats - Watching videos on the food tests to observe colour changes	- Iodine solution - Benedict solution/Clinisti x - Ethanol - Sodium hydroxide, copper sulphate/albusti x
Respiratory system	- describe aerobic and anaerobic respiration - state word equations for aerobic and anaerobic respiration	- Aerobic and anaerobic respiration - Word equations: glucose + oxygen → carbon dioxide + water + energy - glucose → lactic acid + less energy	- Discussing aerobic and anaerobic respiration - Writing down word equation - Carrying out experiments to show release of energy and carbon dioxide from animals and germinating seeds.	- Multimedia - Small animals such as frogs to be returned to their natural environment - Germinating seeds

TRANSPORT SYSTEMS:	describe adaptations of plant leaves to reduce transpiration	Reduction of surface area, thickness of cuticle, distribution of stomata, presence of hairs	- Discussing adaptations of plant leaves to minimize water loss - Carrying out experiment to demonstrate the distribution, role of stomata and water loss - Field touring to identify various types of leaves Avoid plucking leaves from plants	- Potted Plant samples - Potometer Multimedia
	- State the functions of blood - describe the structure of blood vessels - draw and label the structure of blood vessels	- Transport, defence, homeostasis - Veins, arteries and capillaries - Structure of blood vessels - Lumen, valves, thickness	- Discussing the functions of blood - Discussing the structure of blood vessels - Illustrating the structure of veins, capillaries and arteries	- Models - Multimedia

	outline the differences among blood vessels.		- Watching video simulations on the structure of blood vessels - Listing the differences among blood vessels	
REPRODUCTIVE SYSTEMS	- explain asexual reproduction in plants - identify methods of asexual reproduction in plants - state advantages and disadvantages of asexual reproduction - outline the differences between sexual and asexual reproduction in plants	- Asexual/vegetative reproduction - Methods of asexual reproduction: rhizomes, cuttings, tubers - Advantages and disadvantages of asexual reproduction - Differences between sexual and asexual reproduction	- Discussing asexual reproduction in plants - Listing methods of asexual reproduction - Identifying advantages and disadvantages of asexual reproduction - Listing differences between sexual and asexual reproduction	Multimedia
	describe methods of contraception	Natural methods, barrier methods, hormonal methods and spermicides	- Discussing contraceptive methods, - Stating examples	- Pictures - Samples - Multimedia

	advantages and disadvantages of different methods		Identifying advantages and disadvantages	
HEALTH AND DISEASES	- explain the term immunity - describe passive, active, natural and artificial immunity - explain how infants acquire immunity - describe how HIV/AIDS is spread - describe effects of HIV on the body - explain how the spread of HIV/AIDS can be controlled - describe life cycle of malaria parasite - describe methods of controlling malaria	- Immunity - Passive, active, natural and artificial immunity - Breastfeeding, immunisation schedule - HIV/AIDS - Inability to resist infection - Methods of controlling the spread of HIV/AIDS - Life cycle of malaria parasite in humans and in anopheles mosquito - Methods of controlling malaria	- Discussing immunity - Identifying different types of immunity - Discussing how infants acquire immunity - Discussing the spread of HIV/AIDS - Explaining effects of HIV on the body - Discussing methods of controlling the spread of HIV/AIDS - Discussing methods of controlling anopheles	- Resource persons - Multimedia - Resource persons - Multimedia

			mosquito at its different developmental stages	
--	--	--	--	--

8.2.2 CHEMISTRY

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
SEPARATION	- Describe paper chromatography - State the application of paper chromatography	- Paper chromatography: solvent, mixture of dyes, solvent front, initial position of dye and solvent, - Separation of dyes/extracts from plants	Carrying out an experiment to separate mixtures in dyes and plant extracts using paper chromatography	- Filter paper - Thin layer chromatography - Benzene - Toluene - Plant extracts
MATTER	- Describe properties of group I, II, VII and VIII - State the uses of halogens	Physical and chemical properties of group I, II, VII, VIII	Carrying out experiments to react magnesium, iron, zinc and copper, lead with	- Metals - Dilute acids

	- Describe the reactions of metals with water, steam, air and dilute acids - Write equations for the reaction of metals with oxygen, dilute acids and water - List metals in order of decreasing reactivity - Predict the reactivity of a metal from its position in the reactivity series	- Uses of halogens - Reactivity series	air, water/steam and dilute acids	
ACIDS AND BASES	- Identify apparatus used in a titration - Describe an acid /base titration procedure - Carry out acid-base titrations	Acid-base titrations	Titrating dilute sodium hydroxide against hydrochloric acid	- Dilute acids - NaOH(aq) - Phenolphthalein
INDUSTRIAL PROCESSES	- List the raw materials used to manufacture ammonia - Describe the manufacture of ammonia - State the conditions needed for the production of ammonia	- Haber Process - Raw materials: -Hydrogen from electrolysis of water -Nitrogen from fractional distillation of air - Pressure:200 atm	- Describing the Haber process - Conducting educational tours to Sable Chemicals	- Flow chart of the Haber Process - Multimedia - Ammonia solution - Nitric acid - Titration materials

	- State the industrial uses of ammonia - List the raw materials used to manufacture sulphuric acid - Describe the manufacture of sulphuric acid - State the conditions needed for the production of sulphuric acid - State uses of sulphuric acid	- Catalyst: Iron - Temperature: 450 °C – 500 °C - Uses of ammonia: manufacture of ammonium nitrate, dyes - Contact process - Raw materials: sulphur dioxide from burning iron pyrites/sulphur, oxygen from air - Pressure: 1 atm - Catalyst: vanadium (V) Oxide - Temperature: 450 °C – 500 °C - Uses of sulphuric acid: battery acid, plastics, cleaning materials before electroplating	Discussing the contact process	
OXIDATION AND REDUCTION	List alloys of iron	- Alloy formation - mild steel	Discussing the uses of alloys of iron	- Mild steel - Stainless steel utensils

	- State the percentage composition of alloys of iron - Explain the uses and properties of alloys of iron	- stainless steel - cast iron		Cast iron objects
ORGANIC CHEMISTRY	- State the homologous series to which ethanol belongs - Draw the displayed structural formula of ethanol - Describe the production of concentrated ethanol - List uses of ethanol - Define the term global warming - List the causes of global warming	- Ethanol -Fermentation - pH - role of yeast - temperature 30 °C – 35 °C - Fractional distillation - Beverage, medical purpose, fuel, solvent - Global warming - Combustion - Deforestation	- Fermenting sugar solution and maize meal solution (maheu) - Discussing effects of combustion, veld fires and deforestation	- Sugar solution - Maize meal solutions - Yeast/malt - Fractional distillation apparatus - Multimedia

8.2.3 PHYSICS

TOPIC	OBJECTIVES Learners will be able to:	CONTENT (knowledge, skills, values and attitudes)	SUGGESTED ACTIVITIES AND NOTES	SUGGESTED RESOURCES
DATA PRESENTATION	Construct, interpret and analyse pie charts and line graphs	Pie charts and line graphs	Constructing pie charts and line graphs Interpreting and analysing pie charts and line graphs	Graph paper, protractors, compasses, ICT tools
MEASUREMENTS	Express derived quantities in terms of base units	- Derived units - SI units - newton, joule, watt, volts, ampere	Expressing derived quantity units in terms of base units	Print and electronic media

FORCE	- Define pressure - Calculate pressure - Calculate pressure in fluids - Describe effect of depth on pressure - Describe atmospheric pressure - Describe the construction and use of a simple manometer	- Pressure $P=F/A$ - Pressure in fluids $P= h\rho g$ - Atmospheric pressure - manometer	- Determining pressure of solid objects - Demonstrating variation of pressure with depth (refer to structure of dam walls) - Demonstrating atmospheric pressure - Constructing and using a simple manometer to determine fluid pressure	- Solid objects of different cross-sectional area - Container with holes at different depth - Water barometer - Oil, water
	Explain the function and operation of simple fluid systems	Fluid system: siphon and hydraulic systems (car braking system and hydraulic jack)	Demonstrating the operation of a siphon, hydraulic jack and car braking system	Siphon hydraulic jack
	Describe the structures, functions and operations of simple pumps	- Lift pump: Blair pump - Force pump: bicycle pump	Illustrating operation of a Blair pump and a bicycle pump	- Model of a Blair pump - bicycle pump

ENERGY	Describe the functions and designs of a solar cooker and a solar water heater	- Solar cooker - Solar water heater	Discussing qualitatively functions and design of a solar cooker and solar water heater	- A curved reflective surface - Black container - Water heater
TELECOMMUNICATION	- Describe communication over a distance over a distance cell phone and email - List types of media for signal transmission - Describe how signal is transmitted in different media transmitters - Describe the operation of cell phones and related signal transmitters	- Transmission, decoding receiver - Energy conversions - Optic fibre, coaxial cables, WI-FI, sheathed pair cables	- Discussing communication over a distance - Discussion on process of signal transmission in cables and WI-FI - Examining transmitter cables. - Tours to internet service providers and telecommunication companies	ICT gadgets
MAGNETISM	describe hydro and thermal power generation	hydro and thermal power generation	Outlining the energy conversions involved in hydro and thermal power generation	Model hydro and thermal power generations

ELECTRICITY	- Describe electrical hazards and safety precautions - Describe wiring of a three-pin plug - Explain the use of a two-pin plug	- Hazards safety precautions - Hazards: damaged insulation, overheating cables, damp conditions - Safety precautions: earthing, avoid overloading circuits, use insulated cables, do not handle appliances with wet hands, put on rubber foot wear when using appliances - live wire, neutral wire, earth wire, fuse, colour codes	- Discussing the hazards and safety precautions - Wiring a three-pin plug practically - Identifying appliances that use the two-pin plug - Examining appliances that use double insulation - Demonstrating the operation of a fuse and switch	- Charts - Three pin plugs, fuse - Radio - Different fuse ratings - Switches - Three pin plugs - Two pin plugs
--------------------	--	---	---	--

		Two pin plug, double insulation of appliances		
Electricity in the Home	- State uses of electricity in the home - Calculate cost of electricity - Discuss different methods of saving electricity - Explain the use of solar photo voltaic systems	- Heating - Lighting - Powering electrical devices - Meter reading costs per unit - Energy saving bulbs, solar panels, biogas - Alternative sources of energy, switch off appliances not in use - low power rating appliances - solar photo voltaic cells	- Discussing uses of electricity in the home - Reading of electricity meter, determining the costs of electricity used - Discussing ways of saving electricity - Discussing the use of solar systems as alternative sources of energy in homes	- Electrical appliances used in the home e.g. electric jug, iron, welding machines, TV, cake mixers - Energy saving bulbs, other bulbs. Photo voltaic cells

Robot construction and programming	• construct and code	• Robot design and construction	• Constructing robots	• ICT tools • Robotic kits • Resource persons
---	--	---	---	---

9.0 ASSESSMENT

The Heritage-based Combined Science learning area at Lower Secondary school level (Form 1-4) shall be assessed through School Based Continuous Assessment (SBCA) and Summative Assessment (SA). These assessments shall be guided by the principles of inclusivity, practicability, authenticity, transparency, flexibility, validity and reliability. The principles are crucial for creating a supportive and effective learning environment that fosters growth and development in learners at secondary level. In addition to SBCA and SA those aspects that cannot be assessed through the continuous and summative modes will be assessed through learner profiling. The aspect to be profiled will include soft skills among others. Arrangements, accommodations and modifications must be visible to enable candidates with special needs to access assessments. This section covers the assessment objectives, the assessment model, the scheme of assessment, and the specifications for continuous assessment.

9.1 ASSESSMENT OBJECTIVES

The following objectives reflect those aspects of the aims that will be assessed. Specific behavioural learning objectives are stated in each section of the syllabus.

9.1.1 KNOWLEDGE AND COMPREHENSION

Learners should be able to demonstrate knowledge and understanding of:

- scientific instruments and apparatus, techniques and aspects of safety;
- scientific units, terminology, symbols and conventions;
- scientific quantities and how they are determined;
- scientific phenomena, facts and laws, definitions, concepts, theories and models;
- personal, social, economic and environmental implications of science applications.

9.1.2 HANDLING INFORMATION AND SOLVING PROBLEMS

Learners should be able to demonstrate, in familiar and unfamiliar situations, their ability to:

- extract information relevant to a particular context from data presented in diagrammatic, symbolic, graphical, numerical or verbal form;
- use data to recognize patterns, formulate hypotheses and draw conclusions;
- translate information from one form to another;
- communicate logically and concisely;
- explain facts, observations and phenomena in terms of scientific laws, theories and models;

- explain technological applications of science and evaluate their associated personal, social, economic, and environmental implications;
- make logical decisions based on the examination of evidence and arguments;
- apply scientific principles, formulae and methods to solve qualitative and quantitative problems;
- suggest explanations of unfamiliar facts, observations and phenomena;

9.1.3 EXPERIMENTAL SKILLS

Learners should be able to:

- follow instructions for practical work;
- plan, organise and carry out experimental investigations;
- select appropriate apparatus and materials for experimental work;
- use apparatus and materials effectively and safely;
- make accurate, systematic observations and measurements, recognising the variability of experimental measurements;
- observe, measure and record results of experimental procedures;
- identify possible sources of error in experimental procedures;
- draw conclusions and make generalisations from experiments;
- extract information from data presented in diagrammatic, graphical or numerical form.

9.2 Assessment Model

Assessment of learners at Lower secondary school level for Combined Science will be both Continuous and Summative as illustrated in Figure 1. Continuous Assessment will include recorded activities from the School Based Project and other activities done by the learners for assessment while summative will include terminal examinations, end of week, month, term, year or check points assessments.

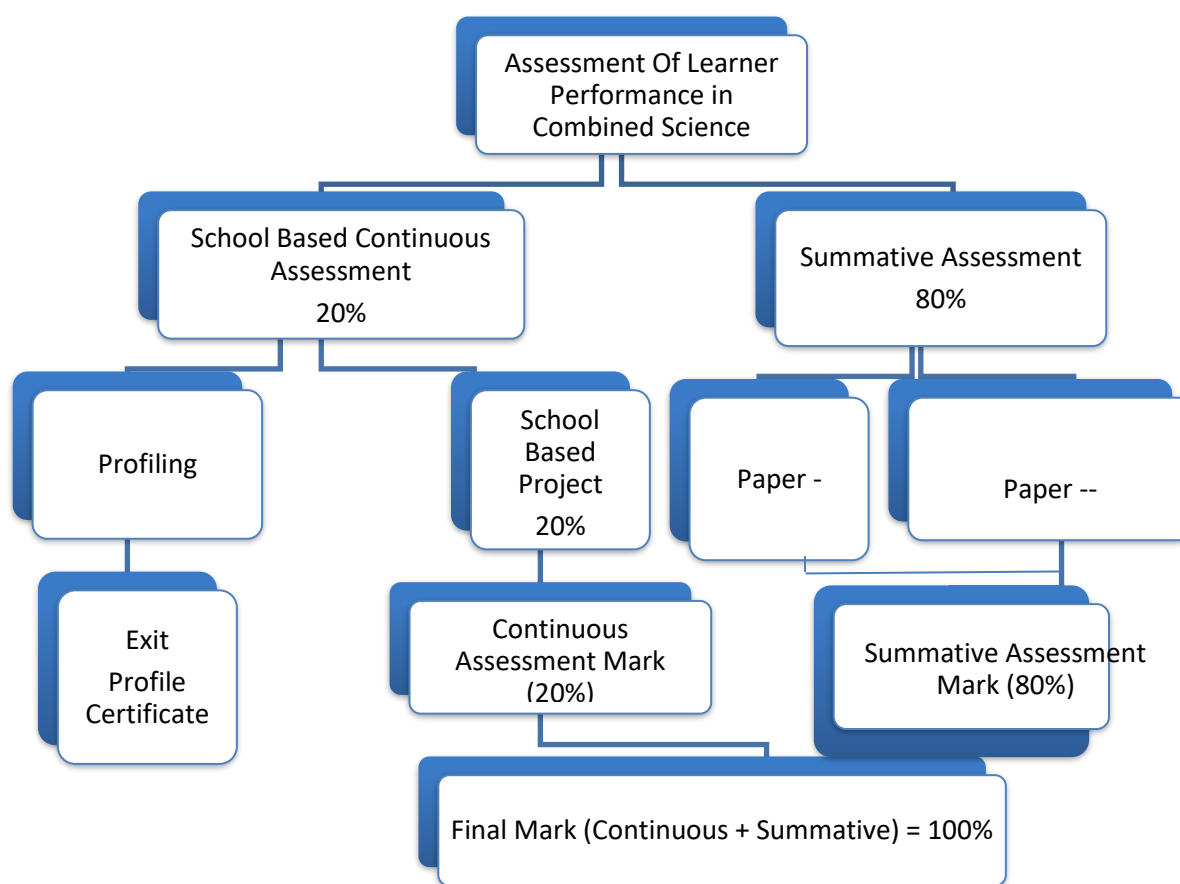


Fig. 1 Assessment Model

In addition, learners shall be profiled and learner profile records established. Learner profile certificates shall be issued for checkpoints assessment in schools as per the dictates of the Teacher's Guide to Learning and Assessment. The aspects to be profiled shall include learner's prior knowledge, values and skills, and subsequently the new competences acquired at any given point.

9.3 Scheme of Assessment

Learners at Lower secondary level will be assessed using both School Based Continuous Assessment and Summative Assessment. From Form 1 -4, learners will do a school-based project per form, per year and per learning area which will contribute 20% to the end of term and year mark. Public examination candidates at secondary level are expected to complete two (2) school-based projects per learning area at form 3 and 4 levels, which will contribute 20% to the final mark at Form 4.

FORM OF ASSESSMENT		WEIGHTING
School Based Continuous Assessment		20%
Summative Assessment		80%
Total		100%

9.3.1 School – Based Project: Continuous Assessment Scheme

The Table given below shows the Learning and Assessment Scheme for the School Based Project.

Project Execution Stages	Project Stage Description	Timelines	Marks
1	Problem Identification	January	5
2	Investigation of related ideas to the problem/innovation	February	10
3	Generation of possible solutions	March	10
4	Selecting the most suitable solution	April-May	5
5	Refinement of selected solution	June	5
6	Presentation of the final solution	July	10
7	Evaluation of the solution and Recommendations	August-September	5
	TOTAL		50

The learning and assessment scheme shows the stages that shall be executed by pupils and the timeline at which each stage shall be carried out. Possible marks, totalling 50, are highlighted to indicate how much can be allocated.

9.3.2 Description of the Summative Assessment

Summative assessment consists of three (3) papers of different weighting

Paper	Paper type	Marks	Duration	Weighting
1	Multiple Choice	40	1 hours	24%
2	Structured questions	100	2 hours	32%
3	Practical	40	1 hour 30 mins	24%
TOTAL				80%

Paper 1

Duration: 1 hour

The paper consists of 40 multiple choice questions. It will be set to cover all topics of the syllabus

Paper 2 Theory (2 hours, 100 marks) section A, B, C and D:

Section A – made up of 6 compulsory structured questions of variable marks which add up to 40 marks.

- Section B 20 marks, will be based on the biology section of the syllabus. It will consist of three free-response Biology questions. Candidates will be required to answer any two questions.
- Section C 20 marks, will be based on the Chemistry section of the syllabus. It will consist of three free-response Chemistry questions. Candidates will be required to answer any two questions

- Section D 20 marks, will be based on the Physics section of the syllabus. It will consist of three free-response Physics questions. Candidates will be required to answer any two questions

Paper 3 Practical Examination (1 hour 30 minutes, 40 marks)

This is a practical consisting of two compulsory questions of 20 marks each from any two sections of the syllabus

NOTE: Examination questions on all papers may be set requiring candidates to apply knowledge to novel situations.

9.4 Specification Grid

Skill	Paper 1	Paper 2	Paper 3
Knowledge	45%	45%	25%
Application and Analysis	40%	40%	50%
Problem solving	15%	15%	25%
TOTAL	100%	100%	100%

9.7 Assessment Instruments/Tools:

The following are suggested tools

Form of Assessment	Suggested Assessment Tools/instruments
Profiling	Check list Observation schedules Rating Scale
Continuous Assessment	Practical activities School based continuous projects

Summative Assessment	Tests Exercises
-----------------------------	------------------------